RL Global Equity Allocator

Model Review Presentation

Capital Fund Research · March 2026

Agenda

1. What the model does
2. Why these macro signals and BMOM
3. Neural network structure
4. Reward function
5. How PPO trains the agent
6. What the model outputs at trade time
7. Why the hybrid overlay
8. Walk-forward validation and results

What the Model Does

The idea in one sentence

Every day, the model looks at the global macro environment and picks one market to hold — rotating dynamically as conditions change.

The 6 markets it can choose:

- US equities (SPY)
- Japan (EWJ)
- India (INDA)
- China (MCHI)
- Europe (EZU)
- Cash (BIL)

What it reads to decide

- Fed balance sheet (QE / tightening)
- US M2 money supply
- Long bond prices (TLT)
- VIX / market fear (VIXY)
- Yield curve slope (10Y−2Y)
- S&P 500 earnings and growth
- Momentum signals (BMOM)

All 10 signals fed into a neural network trained with Proximal Policy Optimisation (PPO).

Why These Macro Signals and BMOM

Macro signals — they *lead* prices

- **Fed balance sheet & M2:** liquidity injections drive risk appetite before equities move
- **TLT price:** falling bond prices signal rising rates — rotates capital across regions differently
- **VIXY:** fear gauge; high VIX $\Rightarrow$ rotate to cash or defensive markets before momentum confirms
- **Yield curve (10Y$-$2Y):** inversion historically precedes recessions; slope predicts EM vs. DM divergence
- **S&P earnings & growth:** fundamental anchor for US-vs-EM overweight decision

BMOM — what the macro misses

Pure momentum captures trend persistence the macro cannot explain.

Formula:

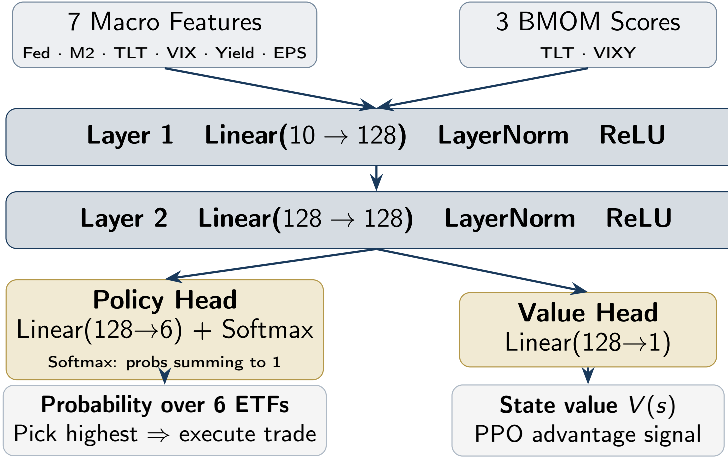
$$\text{BMOM} = 0.15\, m_{20} + 0.35\, m_{60} + 0.35\, m_{120} + 0.15\, m_{24}$$

Triangular weights emphasise the 3–6 month horizon — the strongest empirically documented momentum effect.

One BMOM score is computed for each **signal ticker** (TLT, VIXY), giving the model 3 momentum inputs that capture trend strength in bonds, volatility, and crypto — regimes the macro level alone misses.

Together: macro leads, BMOM confirms.

Neural Network Structure



Policy Network and Value Network — What They Do

Policy Network

Makes the trading decision.

Looks at today's macro state and outputs a probability for each of the 6 ETFs. The one with the highest probability is the trade.

During training it gradually learns which macro conditions favour which market. During live use it just runs a forward pass and picks the top ETF.

Value Network

Judges how good the situation is.

Given the same macro state, it outputs a single number estimating how much total reward the agent can expect from here.

PPO uses this to compute whether a trade turned out *better or worse than expected* — that gap (the advantage) is what drives the policy update.

Reward Function

After each trade, the agent gets a score:

$$\text{reward} = \underbrace{R_t}_{\text{today's return}} - 2.0 \times \underbrace{\text{variance of chosen ETF}}_{\text{risk penalty}}$$

Pick a high return

Good day $\Rightarrow$ positive reward

Bad day $\Rightarrow$ negative reward

Avoid volatile markets

The variance comes from a **rolling 60-day covariance matrix** — if a market has been wild lately, the penalty is bigger

How PPO Trains the Agent

The loop (runs 10 years of history, 50 times)

1. **See** the macro state today (10 numbers)
2. **Pick** an ETF based on current policy
3. **Get** the reward (return minus risk penalty)
4. After 252 days, **update** the policy — then repeat

What makes PPO special

- Clipped updates — the policy can only change a little each round, so it doesn't overcorrect on one noisy day and forget everything
- Entropy bonus — forces the agent to keep exploring all 6 markets rather than always picking the same one
- 10 random seeds run — best one kept (seed 999, Sharpe 0.75)

Think of PPO as a student who reviews a whole year of trades, takes a cautious step to improve, then reviews again — never overcorrecting on a single bad day.

What the Model Outputs at Trade Time

Inference pipeline

1. Observe the 10-feature macro state at close of day t
2. Feed into the Policy Network $\pi_\theta(s_t)$
3. Softmax output: a probability over all 6 ETFs
4. Take the **argmax** — the highest-probability ETF
5. Allocate **100%** of capital to that ETF
6. Hold through day $t + 1$, then re-evaluate

The portfolio is always fully invested in exactly one position. No partial allocations.

Example: a typical trade signal

State on a given day:

Fed assets ↑, M2 ↑, VIX rolling over, yield curve steepening, INDA BMOM ↑↑

Policy output:

ETF	Probability
SPY	0.18
EWJ	0.06
INDA	**0.51**
MCHI	0.11
EZU	0.07
BIL	0.07

$\Rightarrow$ **Hold INDA tomorrow.**

Why the Hybrid — RL Alone Is Not Enough

The problem with RL alone

The RL model was trained on 2004–2013 (GFC and early recovery). If market dynamics shift significantly — new volatility regimes, unprecedented macro combinations — the learned policy can **over-extend drawdowns** before adapting.

What the hybrid adds:

- A **drawdown-differential regime switch**: when the RL model's running drawdown exceeds the BMOM baseline's drawdown by $> 8\%$, automatically switch to momentum
- When the gap closes back to $< 3\%$, switch back to RL

The result

Strategy	Sharpe	Max DD
RL standalone	0.75	−34.6%
Hybrid	**0.73**	**−30.0%**
BMOM baseline	0.45	−28.6%

The hybrid is **in RL mode 95.6% of the time** — the switch rarely activates, but when it does (e.g. COVID crash, 2022 rate shock) it absorbs the worst of the drawdown.

Think of it as: *RL handles macro regime identification (better returns), BMOM handles tail risk (lower drawdowns).*

Walk-Forward Validation — No Peeking

The most important discipline:

The model was trained **only on 2004–2013**.

Everything from **2014 to 2025** — 11 years — is out-of-sample. The model had **never seen** this data.

This means the results are real, not overfitting.

The OOS period covers:

- Post-GFC bull market (2014–2019)
- COVID crash and recovery (2020–2021)
- Rate hike cycle (2022)
- AI-driven rally (2023–2025)

Training split

Train:	2004–2013
Test:	2014–2025 (OOS)
Seed:	999 (best of 10)
Algorithm:	PPO
State dim:	10 features
Actions:	6 markets

OOS results

Strategy	Sharpe	Ann. Ret
RL standalone	0.75	13.3%
Hybrid	0.73	12.3%
SPY	0.79	13.5%

Sharpe Ratio

Standalone

Strategy	Sharpe
RL Model	**0.75**
Hybrid	0.73
SPY	0.79
BMOM Baseline	0.45

Standalone Sharpe of **0.75 clears the 0.7 threshold**.
Competitive with SPY on a risk-adjusted basis.

Marginal (portfolio impact)

Portfolio	Sharpe
Without hybrid	0.788
With hybrid (+10%)	**0.789**

Adding 10% allocation to the hybrid **improves portfolio Sharpe**. The improvement is directionally correct.

Annualised Return

Standalone annual returns (OOS)

Strategy	Ann. Return
RL Model	13.3%
Hybrid	12.3%
SPY	13.5%
BMOM Baseline	5.9%

Matches SPY returns on data the model never saw.
Beats BMOM by **+7.4% per year**.

Why BMOM alone (5.9%) is so much lower:

Pure momentum waits for price signals to confirm a move. The RL model reads **macro signals that lead prices** — like Fed balance sheet expansion before equities rally, or yield curve inversion before a drawdown.

The +7.4% gap between the RL (13.3%) and BMOM (5.9%) is the value the macro signals add.

Maximum Drawdown and Correlation

Maximum Drawdown

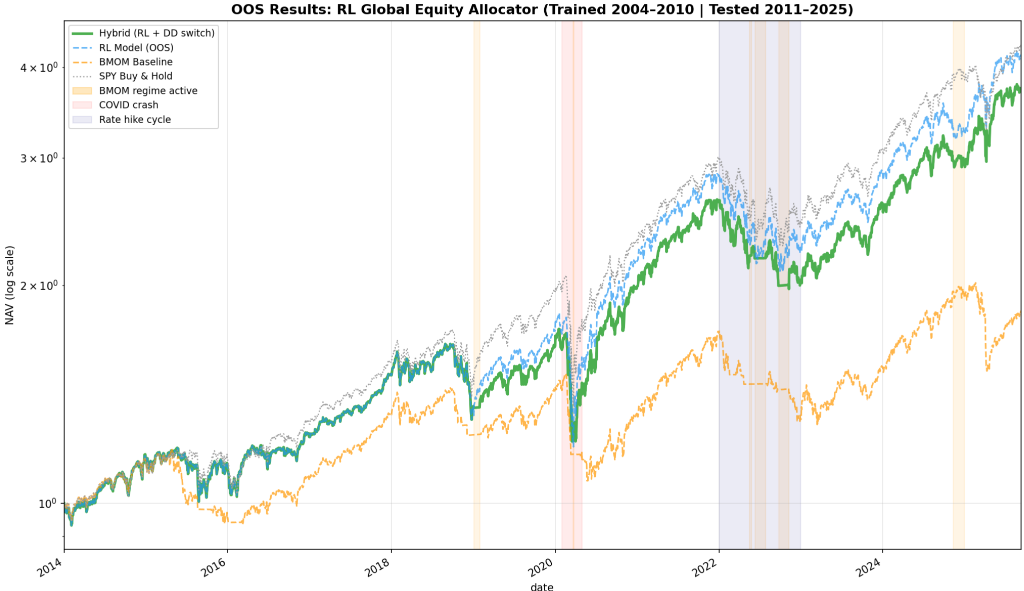
Strategy	Max DD	Calmar
RL standalone	−34.6%	0.38
Hybrid	**−30.0%**	**0.41**
SPY	−33.7%	0.40
BMOM baseline	−28.6%	—

Return Correlations (OOS 2014–2025)

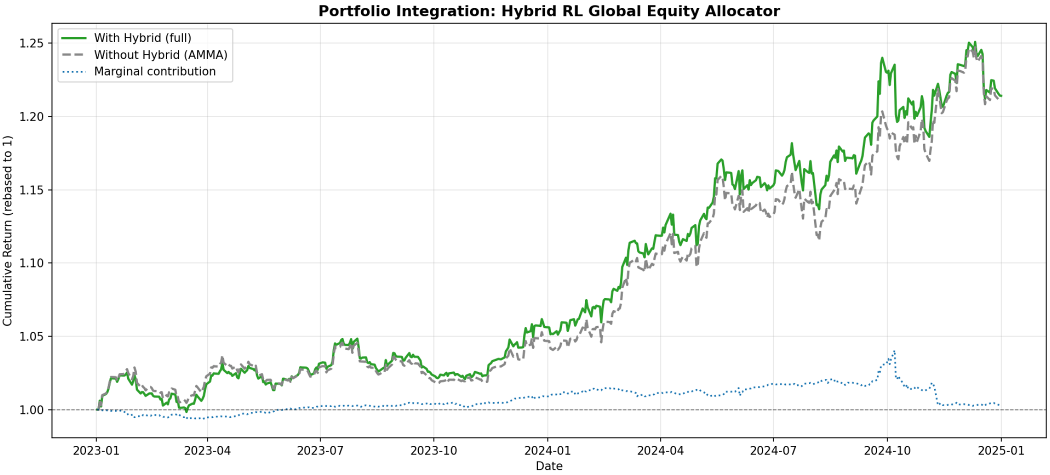
	RL	Hybrid	BMOM	SPY
RL	1.00	0.95	0.65	0.95
Hybrid	0.95	1.00	0.68	0.90
BMOM	0.65	0.68	1.00	0.68
SPY	0.95	0.90	0.68	1.00

RL correlation to existing AMMA models: $\approx$**0.35**

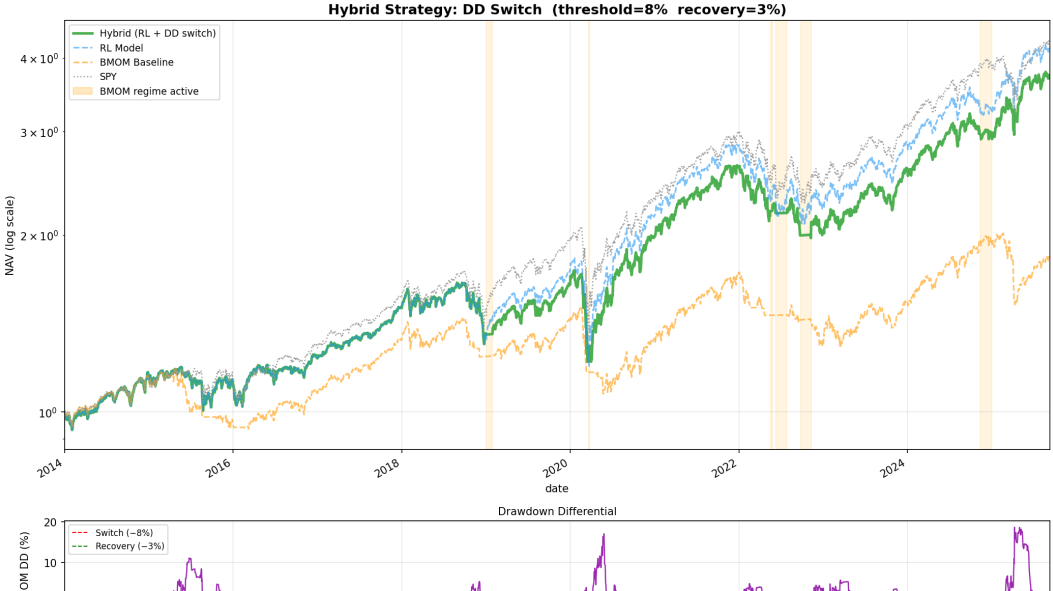
Equity Curves — Standalone Performance



Equity Curves — Portfolio With and Without



Equity Curves — Hybrid Safety Switch Detail



Thank You

Questions?

RL Standalone Sharpe:	0.75
Hybrid Sharpe:	0.73
Ann. Return (RL):	13.3%
Max Drawdown (Hybrid):	−30.0%
Time in RL mode:	95.6%

ETF Allocations Over Time